Math Kangaroo Practice - Grades 3-4

Practice set for 2007 #10

Each original problem is followed by three similar practice problems.

Problem 1: 2007 #10 - Mini-Sudoku

Arranging numbers 1, 2, and 3 in a 3×3 square (Latin square logic)

2007 #10 The square in the figure is a *mini-sudoku*: the numbers 1, 2, and 3 must be written in the cells so that each of them appears in each row and in each column. Harry started to fill in the square. Which number should he write in the cell that is marked by a question mark (?) ?

1	**?**	
2	**1**	

A) only 1 **B)** only 2 **C)** only 3 **D)** 2 or 3 **E)** 1, 2 or 3

Practice 1.1 The square in the figure is a *mini-sudoku*: each row and each column must contain the numbers 1, 2, and 3 exactly once. Leo started to fill in the square as shown. Which number should he write in the cell marked by a question mark (?) ?

?	**2**	
2	**3**	

A) only 1 **B)** only 2 **C)** only 3 **D)** 1 or 2 **E)** 1, 2 or 3

Practice 1.2 Maya started filling in the 3×3 mini-sudoku square using the numbers 1, 2, and 3. Each number must appear once in every row and once in every column. Which number(s) could Maya write in the cell marked by a question mark (?) ?

1	**?**	
2		

A) only 1 **B)** only 2 **C)** only 3 **D)** 2 or 3 **E)** 1, 2 or 3

Practice 1.3 Lucas filled in the mini-sudoku square with numbers 1, 2, and 3 so that each number appears in each row and in each column. Two cells are marked with letters A and B (see picture). What is the sum of the numbers in cells A and B?

1	**A**	
3	**2**	
		B

A) 3 **B)** 6 **C)** 4 **D)** 5 **E)** 7

Answer Key & Explanations

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Solutions & kid-friendly explanations

2007 #10: C In a 3 × 3 mini-sudoku, each row and column must contain the numbers 1, 2, and 3 exactly once. The middle row already has 2 and 1, so its third cell must be 3. The first column has 1 and 2, so its bottom cell must be 3. Now look at the top row: the first cell is 1, so the question mark and top-right cell must be 2 and 3. But column 3 already has 3 in the middle row, so the top-right cell cannot be 3 (it must be 2). Therefore, the question mark cell must be 3 (only 3).

1.1: A The middle row has 2 and 3, so its third cell must be 1. The middle column has 2 and 3, so its bottom cell must be 1. In the top row, the middle cell is 2, so the remaining two cells (the question mark and the top-right cell) must be 1 and 3. If the question mark were 3, the top-right cell would be 1, but column 3 already has 1 in the middle row! Thus, the question mark cannot be 3. It must be 1 (only 1).

1.2: D Column 1 already has 1 and 2, so its bottom cell must be 3. Row 1 has 1 in the first cell, so the remaining cells in the top row must be 2 and 3 in some order. Both completions are valid: if ? is 2, the board can be completed with rows [1, 2, 3], [2, 3, 1], [3, 1, 2]; if ? is 3, the board can be completed with rows [1, 3, 2], [2, 1, 3], [3, 2, 1]. Therefore, Maya can write 2 or 3.

1.3: B The middle row has 3 and 2, so its third cell must be 1. The first column has 1 and 3, so its bottom cell must be 2. In the third column, the middle cell is 1, so the other two cells (top-right and bottom-right B) must be 2 and 3. Since the bottom row already has 2 in the first column, cell B cannot be 2; B must be 3 (making the top-right cell 2). Now in the top row, the numbers are 1, A, and 2, so cell A must be 3. Therefore, the sum of the numbers in cells A and B is 3 + 3 = 6.